Attendance Code: 794664



UNIVERSITY OF LINCOLN

CMP2028M

Natural Language Processing

Week 10: “Machine Translation”

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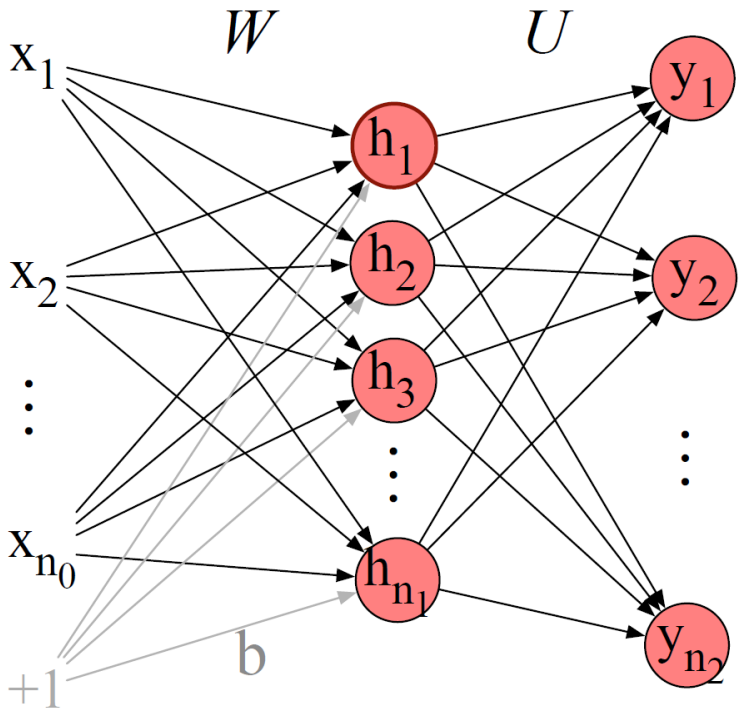
Agenda

- Week 1, 5-6 Feb: Tokenization and regular expressions
- Week 2, 12-13 Feb: N-gram language models
- Week 3, 19-20 Feb: Text classification
- Week 4, 27-27 Feb: Vector semantics and embeddings
- Week 5, 5-6 Mar: Large Language Models (LLMs)
- Week 6, 12-13 Mar: Question answering & information retrieval
- Week 7, 19-20 Mar: Constituency and dependency parsing
- Week 8, 26-27 Mar: Information extraction
- Week 9, 23-24 Apr: Overview of neural networks
- Week 10, 30 Apr-1May: **Machine translation**
- Week 11, 7-8 May: Chatbots and dialogue systems
- Week 12, 14-15 May: Speech recognition and synthesis

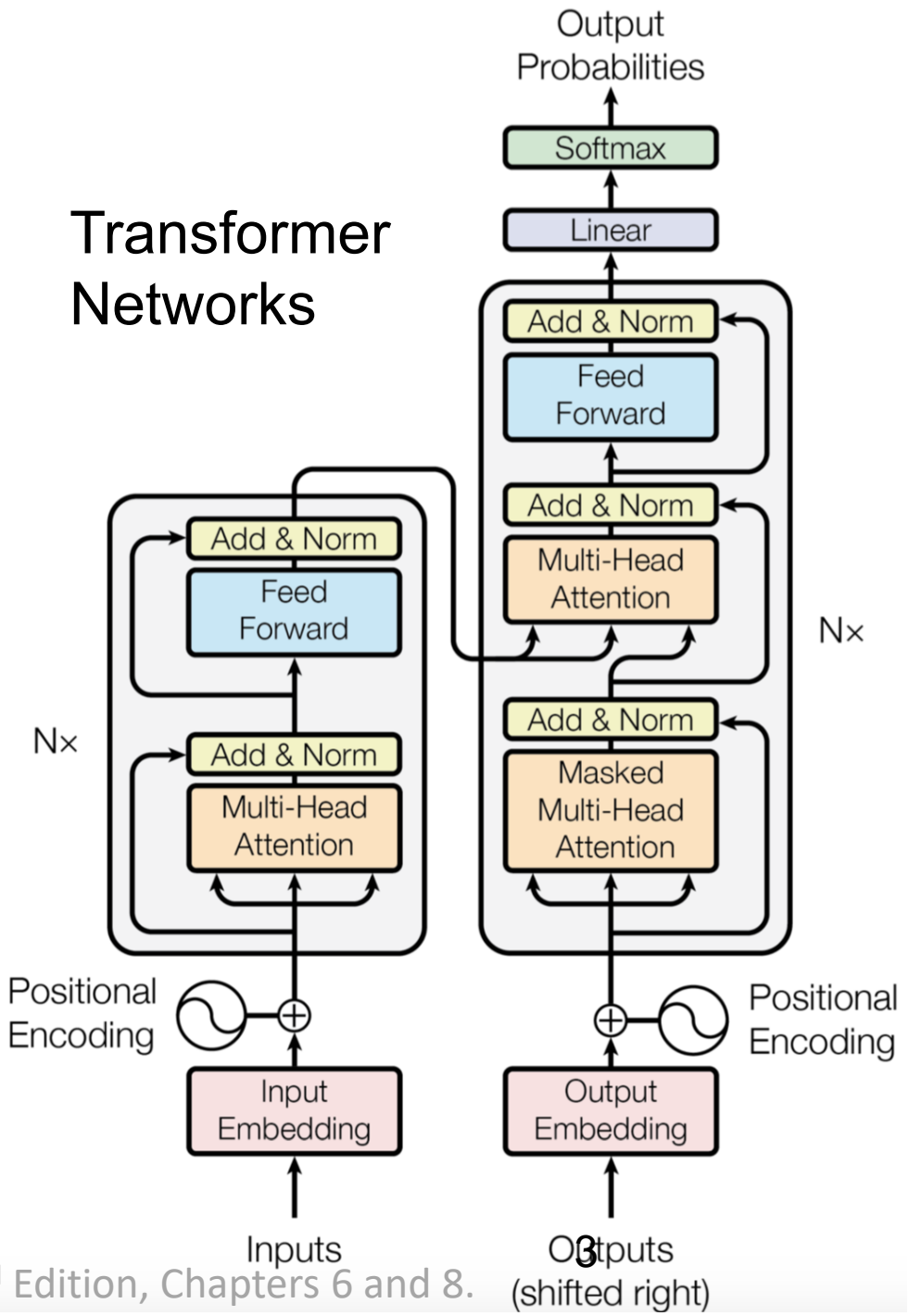
Easter break

Last Week

Multi-Layer Perceptrons (MLPs)



Transformer Networks



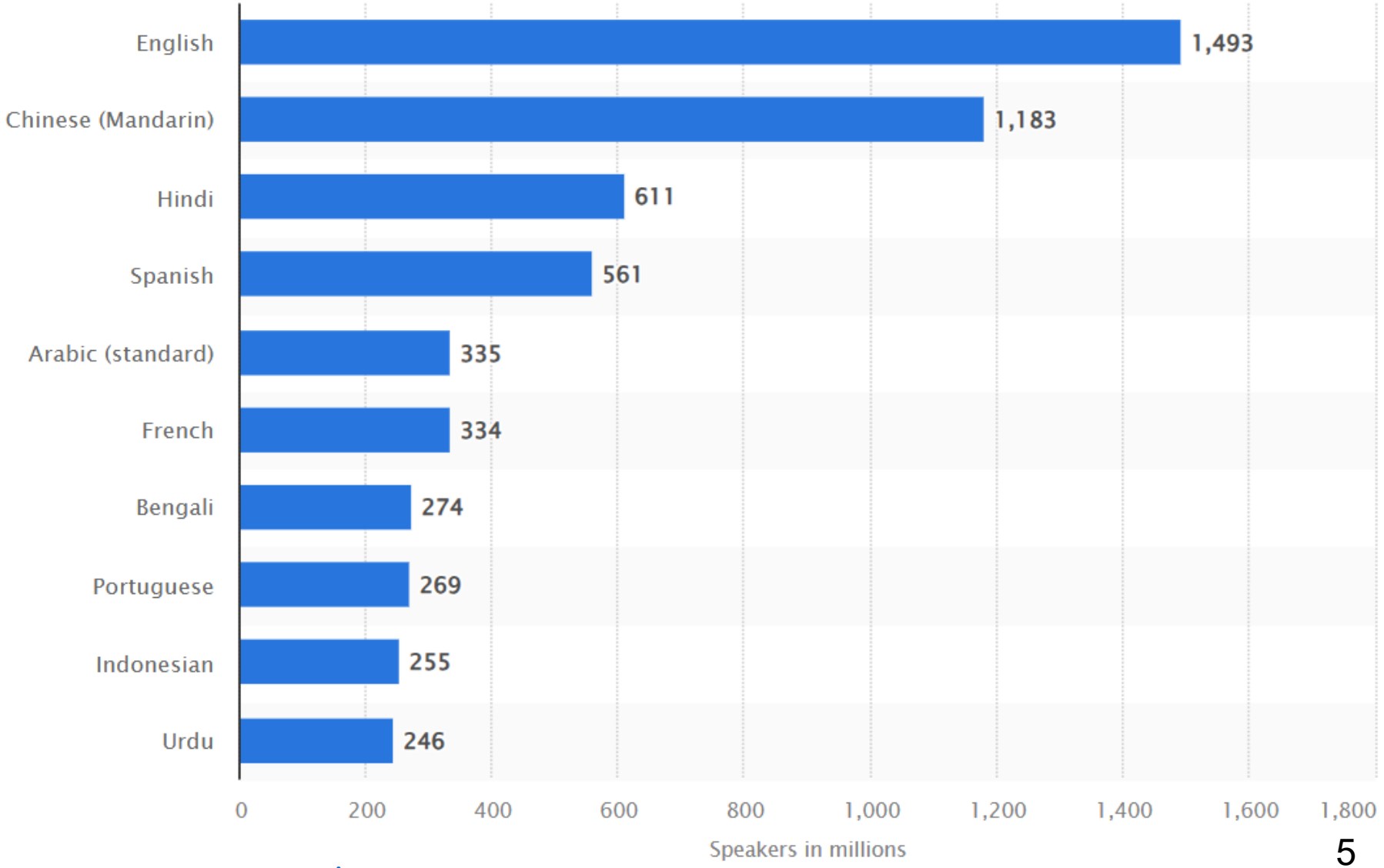
Images from Jurafski and Martin, NLP book, 3rd Edition, Chapters 6 and 8.

Today

- **Motivation and Pre-requisites**
- Machinery
- Evaluation

Acknowledgement: Some images from https://web.stanford.edu/~jurafsky/slp3/12.pdf

Most Spoken Languages Worldwide



source: statista

Motivation

- Communication across languages
  - Enabling people to communicate without human translators.
- Access to information
  - Access to information in other languages than English.
- Commerce and globalisation
  - For contracts, product/customer support, and legal docs.
- Humanitarian and crisis response
  - In conflict/disaster zones rapid translation can be life-saving.
- Cultural preservation
  - For documenting and preserving endangered languages.

Word Ordering in Languages: Examples

Language Types:

Subject Verb Object (SVO)

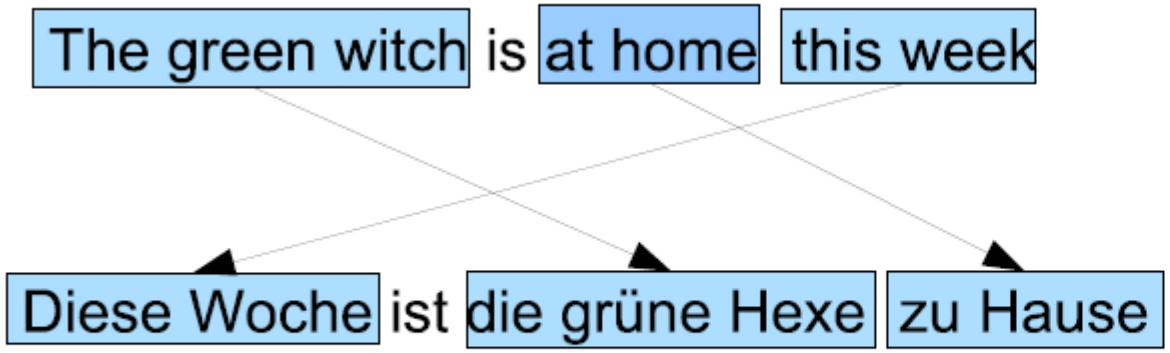
Subject Object Verb (SOV)

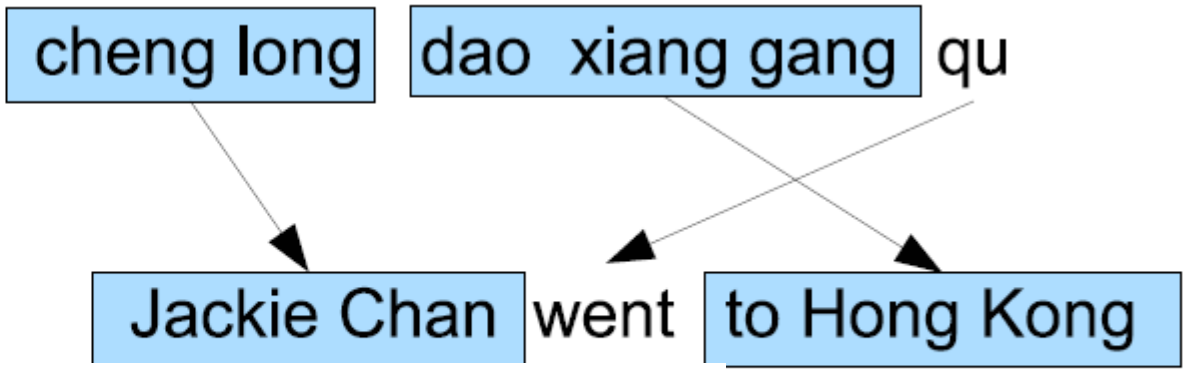
Verb Subject Object (VSO)

English: *He wrote a letter to a friend*

Japanese: *tomodachi ni tegami-o kaita*
friend to letter wrote

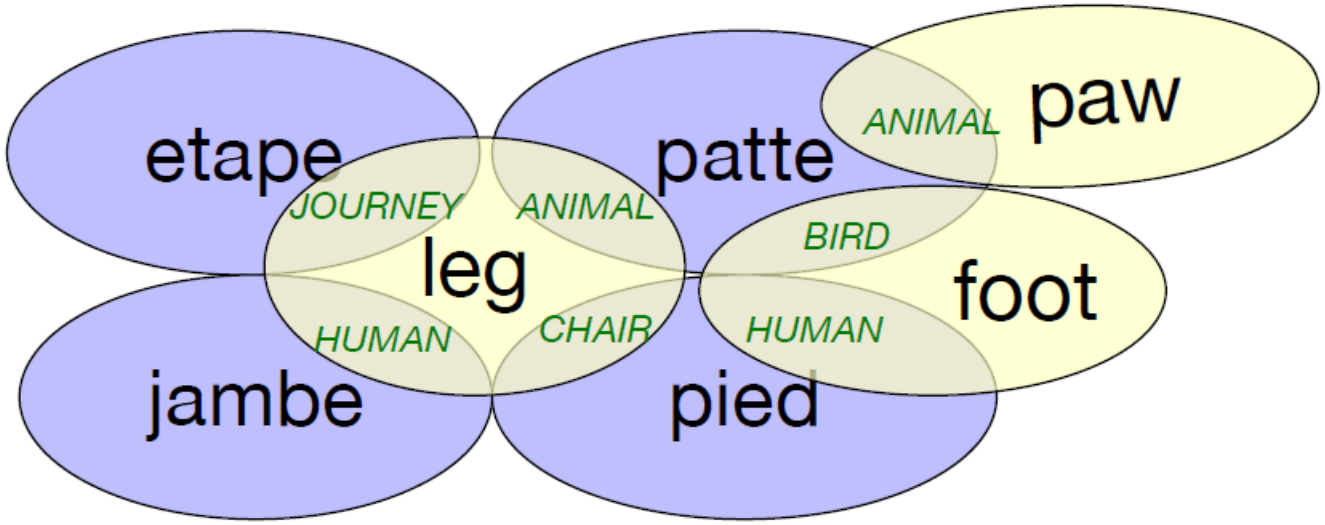
Arabic: *katabt risāla li ṣadq*
wrote letter to friend





Lexical Divergences

For any translation, the appropriate word can vary depending on the context.



In this figure, when **leg** is used about an animal it is translated to French as **patte**; but it is about the leg of a journey, as **etape**; if it is the leg is of a chair, we **pied** is used in French.

Training Data for Translation



- Parallel corpora—sentences in a source language aligned with those in a target language—is needed to train an MT system
- The vast majority of languages do not have parallel corpora.

Tokenisation for Machine Translation (MT)

Original:	corrupted	Original:	Completely preposterous suggestions
BPE:	cor rupted	BPE:	Comple t ely prep ost erous suggest ions
Unigram:	corrupt ed	Unigram:	Complete ly pre post er ous suggestion s

- Subword tokenization (breaking words into smaller parts) can help to represent any word, including those unseen during training.
- Helps to reduce the problem known as ‘Out-of-Vocabulary’ (OOV). The vocabulary is determined automatically by the tokeniser.
- Whilst earlier MT systems used separate two tokenizers (for the source and target languages), modern MT systems use one shared tokenizer across both languages.

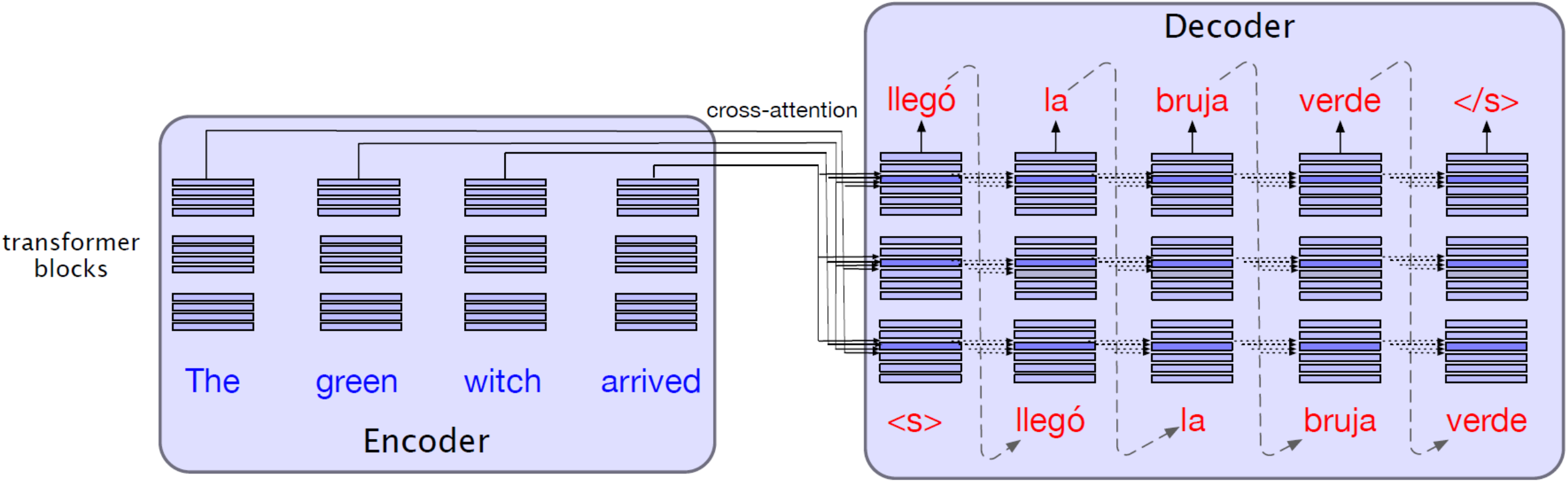
Today

- Motivation and Pre-requisites
- **Machinery**
- Evaluation

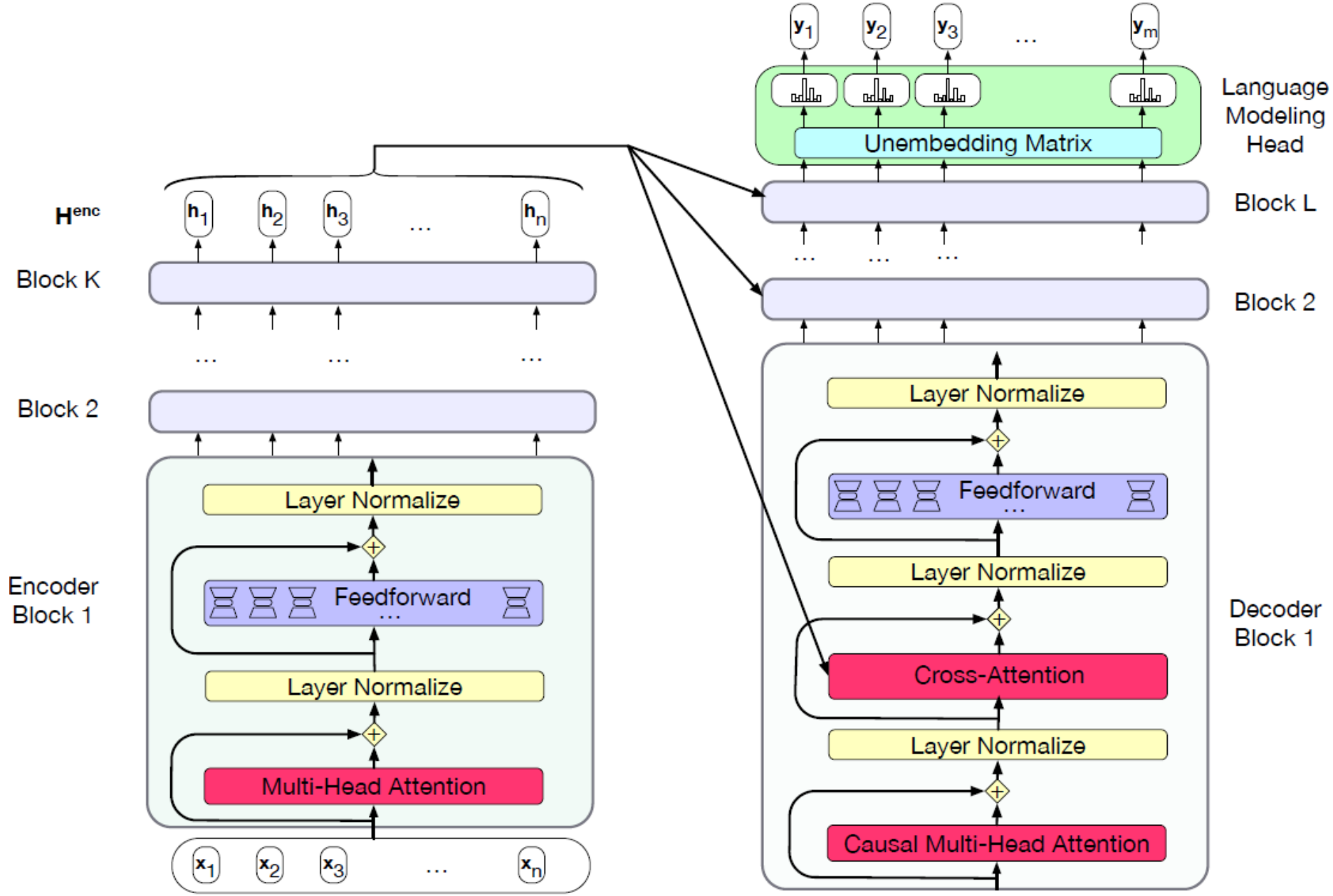
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Neural Machine Translation

Whilst the Encoder performs a limited for of natural language understanding (NLU), the Decoder is responsible for natural language generation (NLG).



Neural Architecture for MT: Encoder-Decoder

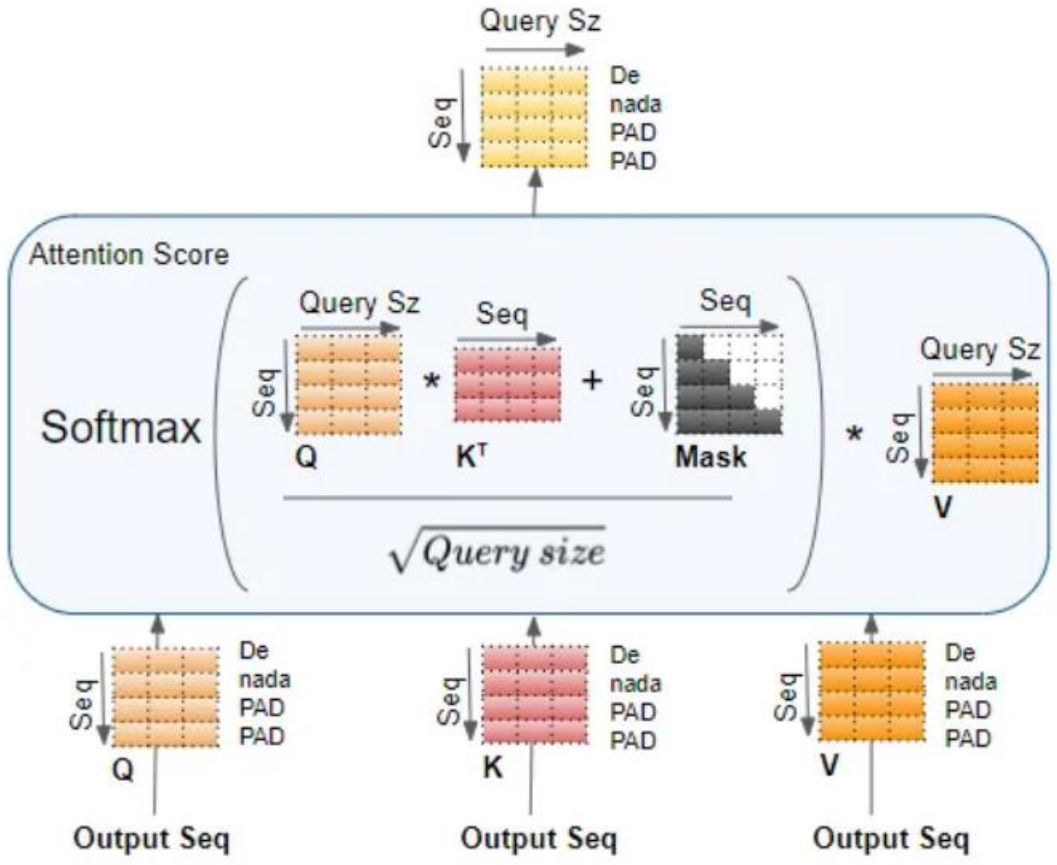


Attention Layers in the Decoder

- **Cross-attention**: "the **queries** come from the previous decoder layer, and the memory **keys** and **values** come from the output of the encoder. This allows every position in the decoder to attend over all positions in the input sequence." – as in the previous figure.

- **Causal Multi-Head Attention**: "self-attention layers allow each position in the decoder to attend to all positions in the decoder up to and including that position. This is implemented inside of scaled dot-product attention by setting to $-\infty$ all values in the input of the softmax which correspond to illegal connections." – the latter are the upper diagonal matrix in the Mask matrix in the next figure.

Causal Self-Attention: Illustration

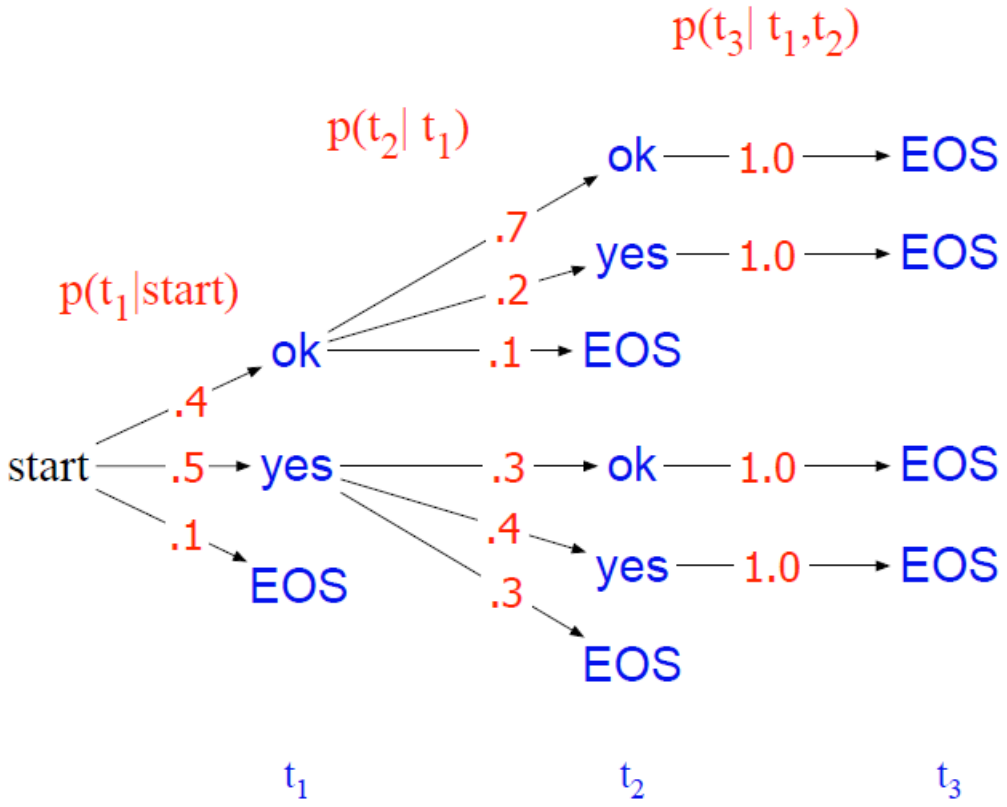


math.exp(float('-inf'))
np.exp(-np.inf)
torch.exp(torch.tensor(float('-inf'))

Example: consider $x = [2, 1, -\infty]$

$$Softmax(x) = \frac{e_i^x}{\sum_j e_j^x} = [0.731, 0.269, 0]$$

The Problem with Greedy Decoding



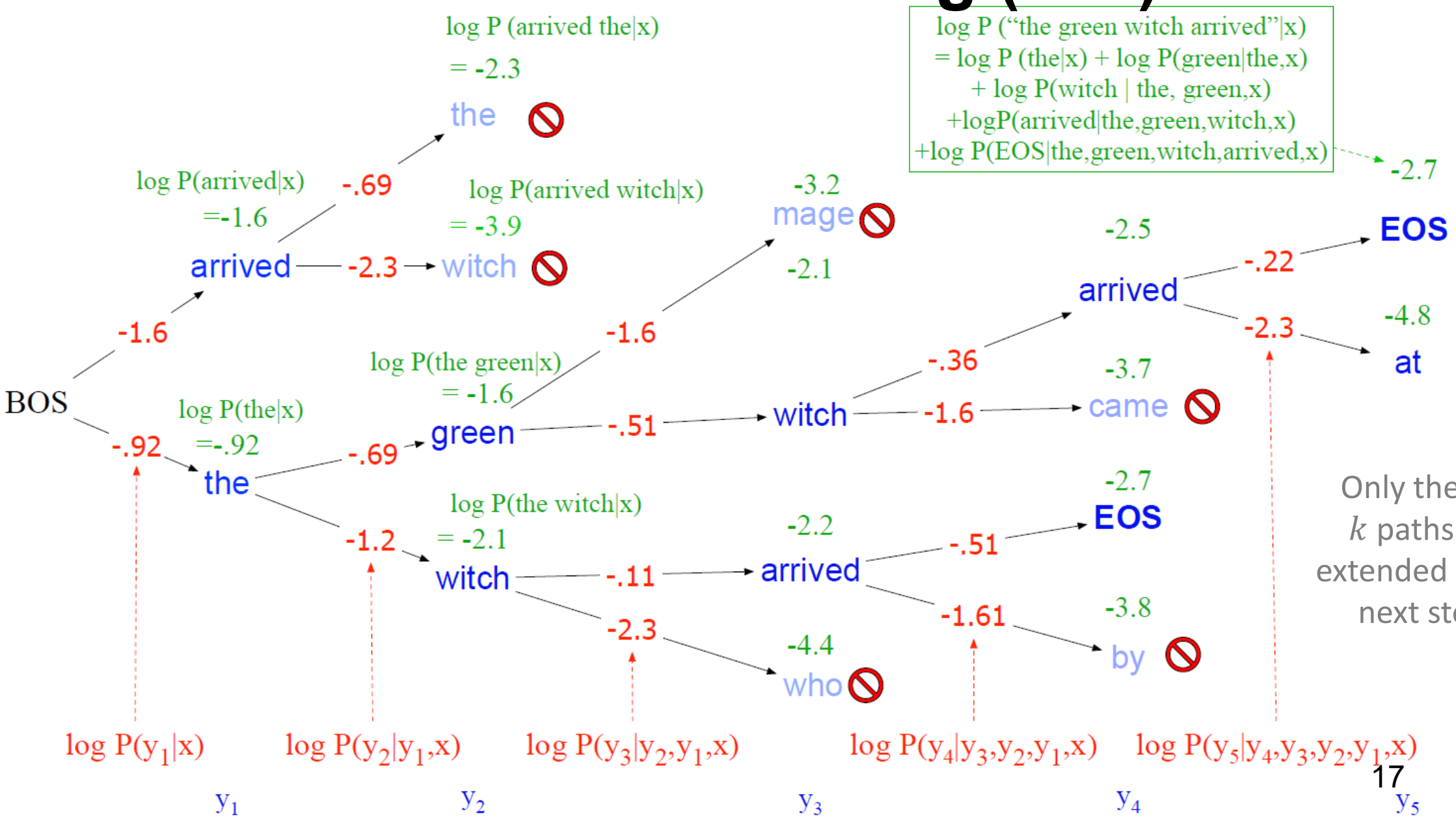
Probability of a sentence:

$$P(w_{1:n}) = \prod_{k=1}^{n} P(w_k|w_{1:k-1})$$

What's the most probable sentence in that search tree?

A search tree showing that greedy decoding chooses tokens yes, yes, EOS, which is not the most probable sequence.

Beam Search Decoding (k=2)



Only the top k paths are extended to the next step!

Normalisation in Sentence Scoring

- We know that we can use the chain rule of probability to score a sentence:

$$\begin{aligned} score(y) &= \log P(y|x) \\ &= \log\left(P(y_1|x)P(y_2|y_1,x)P(y_3|y_1,y_2,x)...P(y_t|y_1,...,y_{t-1},x)\right) \end{aligned}$$

- Because language models generally assign lower probabilities to longer strings, a naive algorithm would choose shorter strings for y. For this reason we can apply length normalisation as follows:

$$score(y) = \frac{1}{t}\log P(y|x) = \frac{1}{t}\sum_{i=1}^{t}\log P(y_i|y_1,...,y_{i-1},x)$$

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MT Systems: Automatic Evaluations

- Character F-Score
  - Measures overlap of character n-grams between prediction and reference.

- BLUE
  - Measures n-gram precision with a penalty for short outputs.

- COMET/BLEURT
  - Learned evaluation metrics using pretrained models.

- BERTScore
  - Measures similarity using contextual embeddings instead of exact matches.

MT Systems: Human Evaluations

- Bilingual raters to ask how intelligible, clear, readable, natural the MT outputs are—in a Likert scale.

- Bilingual experts edit translations, compared against the predicted (machine) outputs to measure quality—based on differences between humans and machine.

- Human evaluations can be considered as more accurate, but they are expensive and time consuming.

What's Next?

- Readings:
  - Jurafsky and Martin. Speech and Language Processing. Chapter 12, 2026. [recommended]
  - Stahlberg, F. "Neural Machine Translation: A Review". Journal of AI Research (JAIR), 2020.

- Workshop **tomorrow**: Machine Translation code.

- Next lecture, in two weeks, check your timetables: Chatbots and Dialogue Systems